

just amounts to coning off a simplicial structure on the boundary of each product cell. This simplicial structure on the product of n copies of X is in fact a Δ -complex structure since the vertices of each of its simplices have a natural ordering given by the dimensions of the cells of which they are barycenters. The action of the symmetric group permuting coordinates respects this Δ -complex structure, taking simplices homeomorphically to simplices, preserving vertex-orderings, so there is an induced Δ -complex structure on the quotient $SP_n(X)$. With the basepoint of X chosen to be a vertex, $SP_n(X)$ is a subcomplex of $SP_{n+1}(X)$ so there is a natural Δ -complex structure on the infinite symmetric product $SP(X)$ as well.

As usual with products, the CW topology on $SP_n(X)$ and $SP(X)$ is in general different from the topology arising from the original definition in terms of product topologies, but one can check that the two topologies have the same compact sets, so the distinction will not matter for our present purposes. For definiteness, we will use the CW topology in what follows, which means restricting X to be a simplicial complex. Since every CW complex is homotopy equivalent to a simplicial complex by Theorem 2C.5, and SP_n and SP are homotopy functors, there is no essential loss of generality in restricting from CW complexes to simplicial complexes.

Here is the main result of this section, the Dold–Thom theorem:

Theorem 4K.6. *The functor $X \mapsto \pi_i SP(X)$ for $i \geq 1$ coincides with the functor $X \mapsto H_i(X; \mathbb{Z})$ on the category of basepointed connected CW complexes.*

In particular this says that $SP(S^n)$ is a $K(\mathbb{Z}, n)$, and more generally that for a Moore space $M(G, n)$, $SP(M(G, n))$ is a $K(G, n)$.

The fact that $SP(X)$ is a commutative, associative H-space with a strict identity element limits its weak homotopy type considerably:

Corollary 4K.7. *A path-connected, commutative, associative H-space with a strict identity element has the weak homotopy type of a product of Eilenberg–MacLane spaces.*

In particular, if X is a connected CW complex then $SP(X)$ is path-connected and has the weak homotopy type of $\prod_n K(H_n(X), n)$. Thus the functor SP essentially reduces to Eilenberg–MacLane spaces.

Proof: Let X be a path-connected, commutative, associative H-space with a strict identity element, and let $G_n = \pi_n(X)$. By Lemma 4.31 there is a map $M(G_n, n) \rightarrow X$ inducing an isomorphism on π_n when $n > 1$ and an isomorphism on H_1 when $n = 1$. We can take these maps to be basepoint-preserving, and then they combine to give a map $\bigvee_n M(G_n, n) \rightarrow X$. The very special H-space structure on X allows us to extend this to a homomorphism $f: SP(\bigvee_n M(G_n, n)) \rightarrow X$. In general, $SP(\bigvee_\alpha X_\alpha)$ can be identified with $\prod_\alpha SP(X_\alpha)$ where this is the ‘weak’ infinite product, the union of the finite products. This, together with the general fact that the map

$\pi_i(X) \rightarrow \pi_i SP(X) = H_i(X; \mathbb{Z})$ induced by the inclusion $X = SP_1(X) \hookrightarrow SP(X)$ is the Hurewicz homomorphism, as we will see at the end of the proof of the Dold-Thom theorem, implies that the map f induces an isomorphism on all homotopy groups. Thus we have a weak homotopy equivalence $\prod_n SP(M(G_n, n)) \rightarrow X$, and as we noted above, $SP(M(G_n, n))$ is a $K(G_n, n)$. Finally, since each factor $SP(M(G_n, n))$ has only one nontrivial homotopy group, the weak infinite product has the same weak homotopy type as the ordinary infinite product. $\square$

The main step in the proof of the theorem will be to show that for a simplicial pair (X, A) with both X and A connected, there is a long exact sequence

$$\cdots \rightarrow \pi_i SP(A) \rightarrow \pi_i SP(X) \rightarrow \pi_i SP(X/A) \rightarrow \pi_{i-1} SP(A) \rightarrow \cdots$$

This would follow if the maps $SP(A) \rightarrow SP(X) \rightarrow SP(X/A)$ formed a fiber bundle or fibration. There is some reason to think this might be true, because all the fibers of the projection $SP(X) \rightarrow SP(X/A)$ are homeomorphic to $SP(A)$. In fact, in terms of the H-space structure on $SP(X)$ as the free abelian monoid generated by X , the fibers are exactly the cosets of the submonoid $SP(A)$. The projection $SP(X) \rightarrow SP(X/A)$, however, fails to have the homotopy lifting property, even the special case of lifting paths. For if x_t , $t \in [0, 1)$, is a path in $X - A$ approaching a point $x_1 = a \in A$ other than the basepoint, then regarding x_t as a path in $SP(X/A)$, any lift to $SP(X)$ would have the form $x_t \alpha_t$, $\alpha_t \in SP(A)$, ending at $x_1 \alpha_1 = a \alpha_1$, a point of $SP(A)$ which is a multiple of a , so in particular there would be no lift ending at the basepoint of $SP(X)$.

What we will show is that the projection $SP(X) \rightarrow SP(X/A)$ has instead the weaker structure of a quasifibration, which is still good enough to deduce a long exact sequence of homotopy groups.

Proof of 4K.6: As we have said, the main step will be to associate a long exact sequence of homotopy groups to each simplicial pair (X, A) with X and A connected. This will be the long exact sequence of homotopy groups coming from the quasifibration $SP(A) \rightarrow SP(X) \rightarrow SP(X/A)$, so the major work will be in verifying the quasifibration property. Since SP is a homotopy functor, we are free to replace (X, A) by a homotopy equivalent pair, so let us replace (X, A) by (M, A) where M is the mapping cylinder of the inclusion $A \hookrightarrow X$. This new pair, which we still call (X, A) , has some slight technical advantages, as we will see later in the proof.

To begin the proof that the projection $p: SP(X) \rightarrow SP(X/A)$ is a quasifibration, let $B_n = SP_n(X/A)$ and $E_n = p^{-1}(B_n)$. Thus E_n consists of those points in $SP(X)$ having at most n coordinates in $X - A$. By Lemma 4K.3(b) it suffices to show that $p: E_n \rightarrow B_n$ is a quasifibration. The proof of the latter fact will be by induction on n , starting with the trivial case $n = 0$ when B_0 is a point. The induction step will consist of showing that p is a quasifibration over a neighborhood of B_{n-1} and over

$B_n - B_{n-1}$, then applying Lemma 4K.3(a). We first tackle the problem of showing the quasifibration property over a neighborhood of B_{n-1} .

Let $f_t: X \rightarrow X$ be a homotopy of the identity map deformation retracting a neighborhood N of A onto A . Since we have replaced the original X by the mapping cylinder of the inclusion $A \hookrightarrow X$, we can take f_t simply to slide points along the segments $\{a\} \times I$ in the mapping cylinder, with $N = A \times [0, 1/2)$. Let $U \subset E_n$ consist of those points having at least one coordinate in N , or in other words, products with at least one factor in N . Thus U is a neighborhood of E_{n-1} in E_n and $p(U)$ is a neighborhood of B_{n-1} in B_n .

The homotopy f_t induces a homotopy $F_t: E_n \rightarrow E_n$ whose restriction to U is a deformation of U into E_{n-1} , where by ‘deformation’ we mean deformation retraction in the weak sense. Since f_t is the identity on A , F_t is the lift of a homotopy $\bar{F}_t: B_n \rightarrow B_n$ which restricts to a deformation of $\bar{U} = p(U)$ into B_{n-1} . We will deduce that the projection $U \rightarrow \bar{U}$ is a quasifibration by using Lemma 4K.3(c). To apply this to the case at hand we need to verify that $F_1: p^{-1}(b) \rightarrow p^{-1}(\bar{F}_1(b))$ is a weak equivalence for all b . Each point $w \in p^{-1}(b)$ is a commuting product of points in X . Let $\widehat{w}$ be the subproduct whose factors are points in $X - A$, so we have $w = \widehat{w}v$ for v a product of points in A . Since f_1 is the identity on A and F_1 is a homomorphism, we have $F_1(w) = F_1(\widehat{w})v$, which can be written $\widehat{F_1(\widehat{w})}v'$ with v' also a product of points in A . If we fix $\widehat{w}$ and let v vary over $SP(A)$, we get all points of $p^{-1}(b)$ exactly once, or in other words, we have $p^{-1}(b)$ expressed as the coset $\widehat{w}SP(A)$. The map $F_1, \widehat{w}v \mapsto \widehat{F_1(\widehat{w})}v'$, takes this coset to the coset $\widehat{F_1(\widehat{w})}SP(A)$ by a map that would be a homeomorphism if the factor v' were not present. Since A is connected, there is a path v'_t from v' to the basepoint, and so by replacing v' with v'_t in the product $\widehat{F_1(\widehat{w})}v'$ we obtain a homotopy from $F_1: p^{-1}(b) \rightarrow p^{-1}(\bar{F}_1(b))$ to a homeomorphism, so this map is a homotopy equivalence, as desired.

It remains to see that p is a quasifibration over $B_n - B_{n-1}$ and over the intersection of this set with $\bar{U}$. The argument will be the same in both cases.

Identifying $B_n - B_{n-1}$ with $SP_n(X - A)$, the projection $p: E_n - E_{n-1} \rightarrow B_n - B_{n-1}$ is the same as the operator $w \mapsto \widehat{w}$. The inclusion $SP_n(X - A) \hookrightarrow E_n - E_{n-1}$ gives a section for $p: E_n - E_{n-1} \rightarrow B_n - B_{n-1}$, so $p_*: \pi_i(E_n - E_{n-1}, p^{-1}(b)) \rightarrow \pi_i(B_n - B_{n-1}, b)$ is surjective. To see that it is also injective, represent an element of its kernel by a map $g: (D^i, \partial D^i) \rightarrow (E_n - E_{n-1}, p^{-1}(b))$. A nullhomotopy of pg gives a homotopy of g changing only its coordinates in $X - A$. This homotopy is through maps $(D^i, \partial D^i) \rightarrow (E_n - E_{n-1}, p^{-1}(b))$, and ends with a map to $p^{-1}(b)$, so the kernel of p_* is trivial. Thus the projection $E_n - E_{n-1} \rightarrow B_n - B_{n-1}$ is a quasifibration, at least if $B_n - B_{n-1}$ is path-connected. But by replacing the original X with the mapping cylinder of the inclusion $A \hookrightarrow X$, we guarantee that $X - A$ is path-connected since it deformation retracts onto X . Hence the space $B_n - B_{n-1} = SP_n(X - A)$ is also path-connected.

This argument works equally well over any open subset of $B_n - B_{n-1}$ that is path-connected, in particular over $U \cap (B_n - B_{n-1})$, so via Lemma 4K.3(a) this finishes the proof that $SP(A) \rightarrow SP(X) \rightarrow SP(X/A)$ is a quasifibration.

Since the homotopy axiom is obvious, this gives us the first two of the three axioms needed for the groups $h_i(X) = \pi_i SP(X)$ to define a reduced homology theory. There remains only the wedge sum axiom, $h_i(\bigvee_\alpha X_\alpha) \approx \bigoplus_\alpha h_i(X_\alpha)$, but this is immediate from the evident fact that $SP(\bigvee_\alpha X_\alpha) = \prod_\alpha SP(X_\alpha)$, where this is the ‘weak’ product, the union of the products of finitely many factors.

The homology theory $h_*(X)$ is defined on the category of connected, basepointed simplicial complexes, with basepoint-preserving maps. The coefficients of this homology theory, the groups $h_i(S^n)$, are the same as for ordinary homology with $\mathbb{Z}$ coefficients since we know this is true for $n = 2$ by the homeomorphism $SP(S^2) \approx \mathbb{C}P^\infty$, and there are isomorphisms $h_i(X) \approx h_{i+1}(\Sigma X)$ in any reduced homology theory. If the homology theory $h_*(X)$ were defined on the category of all simplicial complexes, without basepoints, then Theorem 4.59 would give natural isomorphisms $h_i(X) \approx H_i(X; \mathbb{Z})$ for all X , and the proof would be complete. However, it is easy to achieve this by defining a new homology theory $h'_i(X) = h_{i+1}(\Sigma X)$, since the suspension of an arbitrary complex is connected and the suspension of an arbitrary map is basepoint-preserving, taking the basepoint to be one of the suspension points. Since $h'_i(X)$ is naturally isomorphic to $h_i(X)$ if X is connected, we are done. $\square$

It is worth noting that the map $\pi_i(X) \rightarrow \pi_i SP(X) = H_i(X; \mathbb{Z})$ induced by the inclusion $X = SP_1(X) \hookrightarrow SP(X)$ is the Hurewicz homomorphism. For by definition of the Hurewicz homomorphism and naturality this reduces to the case $X = S^i$, where the map $SP_1(S^i) \hookrightarrow SP(S^i)$ induces on π_i a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$, which one just has to check is an isomorphism, the Hurewicz homomorphism being determined only up to sign. The suspension isomorphism gives a further reduction to the case $i = 1$, where the inclusion $SP_1(S^1) \hookrightarrow SP(S^1)$ is a homotopy equivalence, hence induces an isomorphism on π_1 .

Exercises

1. Show that Corollary 4K.2 remains valid when X and Y are CW complexes and the subspaces U_i and V_i are subcomplexes rather than open sets.
2. Show that a simplicial map $f: K \rightarrow L$ is a homotopy equivalence if $f^{-1}(x)$ is contractible for all $x \in L$. [Consider the cover of L by open stars of simplices and the cover of K by the preimages of these open stars.]
3. Show that $SP_n(I) = \Delta^n$.
4. Show that $SP_2(S^1)$ is a Möbius band, and that this is consistent with the description of $SP_2(S^n)$ as a mapping cone given in Example 4K.5.